Project Design Phase

Proposed Solution Template

Date	27-10-2026
Team ID	
Project Name	Hotel Management and Booking System

Proposed Solution for Hotel Management and Booking System

S. No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Many hotels still rely on manual registers or disconnected tools to manage rooms, reservations, billing, and housekeeping, while guests struggle to check real-time availability and book easily. This leads to double bookings, billing errors, long check-in queues, and poor guest experience.
2	**Idea / Solution Description**	The Hotel Management and Booking System is a full-stack web application (React frontend + Node.js backend) that lets guests search rooms, book online, and pay securely, while staff manage room inventory, check-in/check-out, billing, and housekeeping from one dashboard. Admin controls provide reports, pricing, and user management.
3	**Novelty / Uniqueness**	• Real-time room availability with double-booking prevention • Dynamic pricing based on season, demand, and occupancy • Integrated online payments with automatic invoice generation • Automated email/SMS booking confirmations and reminders • Optional digital check-in and room service request module
4	**Social Impact / Customer Satisfaction**	• Faster, hassle-free booking and check-in improves guest experience • Helps small and mid-size hotels digitize and compete with large chains • Transparent ratings and reviews build guest trust • Reduces paperwork and supports tourism and local employment
5	**Business Model (Revenue Model)**	• Subscription plans for hotels (Basic, Standard, Premium) • Commission on each successful online booking • Premium listing for higher visibility and analytics • Enterprise plans for hotel chains with multi-property management
6	**Scalability of the Solution**	• Expandable to multiple properties, cities, and countries with multi-language and multi-currency support • Mobile-first design with future support via React Native • Modular architecture for easy integration of channel managers, AI-based recommendations, and additional payment gateways